Kshitij Singh Rathore

PhD student in Physics (Thesis Submitted) | Spintronics & Magnonics

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Summary

PhD Research Scholar (thesis submitted; doctorate expected by early October 2026) with over five years of hands-on research experience in the growth, fabrication, and characterization of epitaxial oxide and metal heterostructures for spintronic and magnonic applications. Experienced in a broad range of thin-film deposition, structural, magnetic, and electrical characterization techniques, with practical expertise in instrument operation and laboratory maintenance. Research contributions include 11+ publications (published, submitted, and in preparation), including first-author and collaborative works. Presented research at more than 23 national and international conferences (9 contributed talks and 14 poster presentations) through collaborations with leading research groups in India, USA, Singapore, and Spain. Experienced in mentoring Master's and PhD students in experimental research projects. Currently seeking a postdoctoral position to advance fundamental and applied research in spintronics, magnonics, and magnetic materials while contributing to collaborative, interdisciplinary research.

Education

PhD, Physics – National Institute of Science Education and Research (NISER), Bhubaneswar *2021 – Present*
Advisor: Prof. Subhankar Bedanta (LNMM)
Thesis: *"Ferrimagnetic Insulators for Magnonic and Spintronic Applications"*

Integrated M.Sc. (UG+PG), Physics – Central University of Rajasthan (CURAJ) *2016 – 2021*
CGPA of 8.00/10, and was awarded the **Gold Medal** for the Integrated M.Sc. Physics program.

Higher Secondary (CBSE) – Rawat Public School, Jaipur (**80.40%**) *2014–2015*

Secondary (CBSE) – St. Anselm School, Nagaur (**CGPA: 8.2/10**) *2012–2013*

Experimental & Technical Skills

Sample Growth :
- Pulsed Laser Deposition (PLD)* [APT; Excel Instruments]
- Magnetron Sputtering* [*Mantis, Excel Instruments*]

Structural Characterization :
- X-Ray Diffraction (XRD), Reciprocal Space Mapping (RSM) [Rigaku, SmartLab]
- Reflection High-Energy Electron Diffraction (RHEED) [Staib]
- Transmission Electron Microscopy (TEM) [JEOL]

Magnetic & Spin-Dynamics :
- Ferromagnetic Resonance (FMR) spectroscopy* [NanoOsc]
- Superconducting Quantum Interference Device (SQUID)* [Quantum Design MPMS]
- Physical Property Measurement System (PPMS) [Quantum Design]
- Vector Network Analyzer (VNA) [*Keysight*]
- Magnetic Annealing

Microscopy & Surface :
- Scanning Electron Microscopy (SEM) [*ZEISS*]
- Optical Profilometer

Device Fabrication :
- Photolithography [Heidelberg Instruments]
- Spin Coating
- Reactive ion etching (RIE)
- Wire Bonding

Computational and Data tool : Origin, LaTeX, MATLAB, Python, Fortran, C/C++, SolidWorks, HTML, 3D printing

Current Instrument Development:
- Time-Resolved MOKE (TRMOKE), *middle stage of development*
- Spin-Torque Ferromagnetic Resonance (ST-FMR), setup complete; data testing
- Harmonic Hall Voltage Measurement - setup complete; result analysis stage

*Beyond routine operation, hands-on experience in the installation, calibration, troubleshooting, and preventive maintenance of these instruments, including servicing carried out jointly with the manufacturer's engineers (appreciation certificate from the manufacturing company available).

Research Experience

Doctoral Researcher – Spintronics & Magnetic Materials Group *Jan 2021 – Present*
NISER, Bhubaneswar
Advisor: Prof. Subhankar Bedanta (LNMM)
Investigating exchange-coupled ferrimagnetic garnet heterostructures for coherent spin-current transport and hybrid magnonic applications.

Thesis chapters comprise:

- Comparative spin pumping study in YIG/CoFeB and YIG/Py heterostructures, identifying YIG/CoFeB as the cleaner platform for interfacial spin transport.
- Ditaction of exchange-coupled hybrid magnon modes in YIG/CoFeB/Cu, enhancing spin-mixing conductance and spin pumping efficiency at low temperature.
- Growth optimization of epitaxial YIG/GdIG bilayers, linking substrate choice and growth sequence to strain relaxation and interfacial quality.
- Enhanced spin pumping and spin-to-charge conversion in coupled YIG/TmIG/Pt garnet heterostructures.
- Observation of coherent magnon–magnon coupling and hybridized resonance modes in YIG/TmIG bilayers.

Master's Thesis: Transparent Conducting Oxide thin films of AZO *Jan – Jul 2021*
Central University of Rajasthan
Advisor: Dr. Yugandhar Bitla

- Developed a cost-effective spin-coating route for Aluminum-doped Zinc Oxide (AZO) transparent conducting oxide thin films and evaluated their structural and electrical properties on Quartz and Mica.
- Also got training to use Arc Melting, High temperature furnace, Bulk sample preparation, Thin film sol-gel technique.

Bachelors Minor Project: Plasma Physics *Jan – May 2019*
Central University of Rajasthan
Advisor: Dr. Sukhmander Singh

- Study of the motion of charge particle in electric and magnetic fields for fusion devices like tokamaks.

Bachelors mini Project: Optical Tweezer *DEC 2019*
Central University of Rajasthan
Advisor: Dr. Brijesh Kumar Singh

- Received the third Prize for Best Poster in International Conference on Light and Photonics.

Research Collaborations & Training

- Trained in with Prof. Vivek Mallik at **IIT Roorkee** for reciprocal space mapping (RSM) in typimanganite-basedased films.
- Trained in with Dr. Jose Santiso at **ICN2, Spain**, via the NFFA project, on PLD-based sample fabrication and characterization (XRD, FIB lamella preparation, aberration-corrected TEM).
- Collaborated with Dr. Anjana Dogra at **CSIR-National Physical Laboratory (NPL)**, New Delhi, on joint sample preparation.
- Trained and contributed to writing funded research proposals under DST-SERB, I-Hub, National Quantum Mission (NQM), ANRF, NFFA-Europe, and DAE.
- Additional facilities available within the laboratory such as Atomic Force Microscopy (AFM), Magnetic Force Microscopy (AFM), Magneto-Optical Kerr Microscopy (MOKE), MuMax3 micromagnetic simulations, glove-box processing, Focused Ion Beam (FIB) milling, Raman spectroscopy, thermal evaporation, Low-Energy Electron Diffraction (LEED), and UV–Visible spectroscopy. Although these techniques were not part of my primary research, I am familiar with their working principles and applications. However have not readily acquire hands-on expertise as required.
- Possess practical knowledge of the operation, troubleshooting, and routine maintenance of high-vacuum systems, including scroll pumps, turbomolecular pumps, vacuum gauges, gas handling lines, and associated vacuum hardware.
- In addition to my primary research in magnonics and spin pumping, working in a multidisciplinary spintronics laboratory has provided broad exposure to the fundamental concepts of skyrmionics, current-driven domain wall dynamics, organic spintronics, antiferromagnetic spintronics, orbital Hall effect, spin–orbit torque, and flexible spintronics across a wide range of magnetic material systems, strengthening my understanding of these research areas.
- Mentored 5 M.Sc. thesis students at NISER: Mr. Abhishek Singh (2024; now PhD, Helmholtz-Zentrum Dresden-Rossendorf, Germany); Mr. Gaurav Kanu (2023; now PhD, University of Duisburg-Essen, Germany); Mr. Santosh Sourav Sahoo (2023; now Diploma in Radiological Physics, BARC); Mr. Akilan K (2022; now PhD, University of Lorraine, France); and Mr. Siba Prasad (2026; M.Sc., Central University of Hyderabad).
- Currently mentoring one ongoing M.Sc. student, Mr. Jiten, and training two PhD students.
- Guided summer/short-term project students: Tapasmini Beura, Teena Meena, Rojalin Swain, and Kuldeep.

Publications

Published (peer-reviewed)

1. **K.S. Rathore**, A. Swain, J. Lim, A. Mishra, P. Gupta, Y.H. Lee, J. Luwen, A. Hoffmann, R. Mahendiran, S. Bedanta, *Phys. Rev. Applied* **25**, 024036 (2026).
2. A. Mishra, **K.S. Rathore**, S.P. Mahanta, S. Bedanta, *Phys. Rev. B* **113**, 014407 (2026).
3. A. Swain*, **K.S. Rathore***, P. Gupta, A. Mishra, Y.H. Lee, J. Lim, A. Hoffmann, R. Mahendiran, S. Bedanta, *Appl. Phys. Lett.* **125**, 012406 (2024).
4. A. Mishra, P. Das, R. Chhatoi, S. Dash, S. Sahoo, **K.S. Rathore**, P.R. Cha, S.-C. Lee, S. Bhattacharjee, S. Bedanta, *Phys. Rev. Applied* **24**, 064032 (2024).

Submitted / In preparation

1. **K.S. Rathore**, A. Mishra, S.P. Mahanta, S. Sahoo, A. Swain, J. Santiso, S. Bedanta, *"Interface-resolved structural properties of epitaxial $Y_3Fe_5O_{12}/Gd_3Fe_5O_{12}$ bilayers grown on GGG(111) by pulsed laser deposition"* – Submitted (2026)
2. S.P. Mahanta, M.S. Pawar, **K.S. Rathore**, M. Swain, A. Mishra, A. Ghosh, S. Bedanta, *"Impact of Graphene Underlayer on Magnetization Dynamics of NiFe Thin Films"* – Under Review
3. P. Gupta, A. Mishra, A. Sahoo, **K.S. Rathore**, A. Swain, S. Sahoo, S. Dash, M. Kuila, A. Barman, A. Azevedo, A. Haldar, A.A. Tulapurkar, A. Hoffmann, C. Murapaka, D. Atkinson, H. Ding, M. Kläui, P.K. Muduli, P.S. Anil Kumar, S. Bedanta, *"Spin to charge interconversion with heavy metals, topological insulators, transition metal oxides, transition metal dichalcogenides and Rashba-like states"* – Under Review
4. S. Satapathy, **K.S. Rathore**, S.P. Mahanta, Km. Alka, H.K. Singh, S. Bedanta, K.K. Maurya, *"Observation of the Inverse Spin Hall Effect in RF-Sputtered YIG/$SrRuO_3$ All-Oxide Heterostructures"* – Under Review
5. M. Swain, S.P. Mahanta, **K.S. Rathore**, A. Sahoo, S. Bedanta, *"Enhanced Spin-to-Charge Conversion in $Co_{40}Fe_{40}B_{20}/\beta$-W/MnPc Heterostructures via MnPc Interfacial Engineering"* – In preparation
6. S.P. Mahanta, A. Sahoo, M. Swain, **K.S. Rathore**, A. Mishra, L.N. Serkovic-Loli, L. Aviles Felix, S. Bedanta, *"Influence of Graphene and Antimonene Underlayers on Magnetization Dynamics in FeCo Thin Films"* – In preparation
7. **K.S. Rathore** *et al.*, *"Tailoring Spin Pumping in $Y_3Fe_5O_{12}$/Metallic-Ferromagnet Heterostructures"* – In preparation
8. **K.S. Rathore** *et al.*, *"Exchange-Coupled Hybrid Modes in $Y_3Fe_5O_{12}$/CoFeB/Cu Heterostructures"* – In preparation
9. **K.S. Rathore** *et al.*, *"Structural and Magnetic Properties of GGG- and sGGG-Based $Y_3Fe_5O_{12}/Gd_3Fe_5O_{12}$ Heterostructures"* – In preparation

Conference Presentations & Awards

Contributed Talks (9)

1. Presented a contributed talk at the IEEE Magnetics Society Italy Chapter – PETASPIN School on "Spintronics: Fundamentals and Applications," Messina, Italy (Dec 2022).
2. Presented a contributed talk (virtual) at the Around-the-Clock Around-the-Globe (AtC-AtG) Magnetics Conference (Oct 2024).
3. Presented a contributed talk at the International Conference on Magnetic Materials & Applications (ICMAGMA 2025), IISc Bangalore (Feb 2025).
4. Presented a contributed talk at the Material Research Conclave (MRC), IMMT Bhubaneswar (Sep 2025).
5. Presented a contributed talk (virtual) at the Around-the-Clock Around-the-Globe (AtC-AtG) Magnetics Conference (Nov 2025).
6. Presented a contributed talk at Smart and Novel Material for Next Generation (SNMNG), SOA University, Bhubaneswar (Dec 2025).
7. Presented a contributed talk at the Symposium in Magnetism and Spintronics (SMS-3), IIT Hyderabad (Jan 2026).
8. Delivered a lightning talk at the IEEE International Magnetics Conference (INTERMAG 2026), Manchester, UK (Apr 2026).
9. Presented a contributed talk at the IEEE International Magnetics Conference (INTERMAG 2026), Manchester, UK (Apr 2026).

Poster Presentations (14)

1. Presented a poster (virtual) at the Around-the-Clock Around-the-Globe (AtC-AtG) Magnetics Conference (Aug 2022).
2. Presented a poster (virtual) at the Indo-Japan Workshop on Interface Phenomena for Spintronics (IJW-IPS 2022) (Mar 2022).
3. Presented a poster (virtual) at the Symposium in Magnetism and Spintronics (SMS-1) (Mar 2022).
4. Presented a poster at the Symposium on Interdisciplinary Sciences (SIS-I), NISER Bhubaneswar (Jan 2023).
5. Presented a poster (virtual) at the Around-the-Clock Around-the-Globe (AtC-AtG) Magnetics Conference (Sep 2023).
6. Attended the Physics of Quantum Matter School, NISER Bhubaneswar (May 2023).
7. Presented a poster at the 68th DAE Solid State Physics Symposium (DAE SSPS), GITAM Visakhapatnam (Dec 2024).
8. Presented a poster at the Symposium in Magnetism and Spintronics (SMS-2), IIT Bombay (Jul 2024).
9. Presented a poster at the Indo-German Kick-off Workshop (IGW-2024), NISER Bhubaneswar & Sterling Puri (Nov 2024).
10. Presented a poster at the DAE-BRNS 11th Theme Meeting on Ultrafast Sciences (UFS-2024), NISER Bhubaneswar (Nov 2024).
11. Presented a poster at the Quantum Condensed Matter Symposium 2024, IIT Guwahati (Dec 2024).

12. Presented a poster at the Workshop on Processability and Applications of Reticular Advanced Materials (PARAM-25), NISER Bhubaneswar (Jan 2025).
13. Presented a poster at the Symposium on Interdisciplinary Sciences (SIS-II), NISER Bhubaneswar (Nov 2025).
14. Presented a poster at the 2026 International Symposium on Integrated Magnetics (iSIM 2026), Manchester, UK (Apr 2026).

Awards & Fellowships exam qualification

- Received the **iSIM Student Award**, 2026 ($300)
- Received the **IEEE Travel Grant Award**, 2026 ($1000).
- Qualified the Graduate Aptitude Test in Engineering (GATE) in Physics, 2021, with All India Rank (AIR) 804.
- Qualified the Joint Entrance Screening Test (JEST), 2021, with All India Rank (AIR) 503.
- Received the DST INSPIRE Fellowship.
- Qualified the Joint Admission Test for M.Sc. (IIT-JAM) in Physics, 2019, with All India Rank (AIR) 1204.
- Qualified the Joint Entrance Examination (IIT-JEE), 2016.
- Awarded Gold Medal for the Integrated M.Sc. Physics program, Central University of Rajasthan.
- Awarded a CSIR-Lab Summer Research Fellowship for an optical-tweezer-based COVID-19 enzyme-detection project in May 2021.
- Qualified the Central Universities Common Entrance Test (CUCET) 2016 for admission to the Integrated M.Sc. program.
- Secured **State Rank 26** and **International Rank 808** in the International Mathematics Olympiad (IMO), 2014.
- Received the **Academic Excellence Award** throughout schooling up to Higher Secondary level for academic performance.

Miscellaneous

Teaching Experience

- Served as a Teaching Assistant at NISER Bhubaneswar for more than 200 students across three semesters, teaching P102 (**Introductory Physics II**), P141 (**General Physics Laboratory II**), and P142 (**General Physics Laboratory I**).
- Taught Physics to Class XI and XII students for eight months at the higher secondary school level, delivering classroom instruction and mentoring students in preparation for board examinations.

Leadership & Achievements

- Served as the Department Representative for the Department of Physics, Central University of Rajasthan (2016–2017).
- Awarded the **"Student of the Year"** at Rawat Public School, Jaipur (2015).
- Served as the **Head Boy** at Rawat Public School, Jaipur (2015).
- Served as the **House Captain** at St. Anselm School, Nagaur (2013).
- Served as the **Student Representative** of the Volleyball Club, Central University of Rajasthan (2016–2017).
- Awarded **First Prize** in the National Science Day Quiz Competition held on 20 February 2019.
- Attended and organized numerous national and international seminars, webinars, workshops, and conferences on topics including the evolution of quantum theory, magnetism and spintronics, quantum entanglement, skyrmions, magnonics, and other emerging areas of condensed matter physics.
- Served as a **Mega Mess Committee Member** at the Central University of Rajasthan(2017–2020) and at NISER (2022–2025).

Sports & Extracurricular Activities

- National-level Volleyball Player.
- Represented the university in the **All India West Zone Inter-University Volleyball Championship** (2017, 2018, and 2019).
- Won the **Gold Medal** in the 2nd Annual Volleyball Festival held at the Central University of Rajasthan (2018).
- Won the **Gold Medal** in the University Volleyball Championship (2018 and 2019).
- Won the **Silver Medal** in Volleyball and Handball at the Intra-Central University of Rajasthan Sports Meet (2019).
- Secured the **Silver Medal** at the All India Inter-IISER Sports Meet (IISM 2025).
- Won the **Gold Medal** in Cricket at the Intra-Central University of Rajasthan Sports Meet (2016).
- Won the **Bronze Medal** in the 11 km Marathon organized by the Adventure Sports Club, Central University of Rajasthan.

Personal Data

- **Date of Birth:** 28 March 1997
- **Time of Birth:** 2:02 PM
- **Height:** 6 ft 0 in (183 cm)
- **Weight:** 80 kg
- **Blood Group:** A^+
- **Father's Name:** Mr. Renwat Singh Rathore (Government Teacher)
- **Mother's Name:** Mrs. Nilesh Kanwar (Homemaker)
- **Brother's Name:** Dr. Yashovardhan Singh Rathore (Orthopedics, Ajmer Government Hospital)
- **Permanent Address:**
  Renwat Singh Rathore,
  Village–Sudary, Post–Banwarla, Via–Ren,
  Tehsil–Degana, Nagaur, Rajasthan–341514, India.
- **Social Media Profiles:** LinkedIn | Facebook | Instagram | X